

Assignment : 1

Q-1

a Differentiate between technical and non technical problem with example.

Ans	Technical problem	Non technical problem
→	It requires technical knowledge, tools or computational method to solve it.	→ It may involve human, social, emotional factors.
→	Usually has clearly defined inputs, processes & outputs.	→ May not have a single clearly defined sol ⁿ .
→	Can be solved using algorithm, engineering method.	→ Often requires communication, decision making or behavioral change.
→	Ex - Calculating the shortest path between two cities on a map.	→ Ex - Deciding how to resolve a conflict between two employees in an office.

b What is mathematical thinking? How is it different from design thinking?

Ans Mathematical thinking is a way of solving using logic, reasoning, patterns, calculations and

mathematical principles.

While

"Design thinking is a human centered approach to solving problems by understanding users, identifying their needs, generating ideas, creating prototypes & testing solⁿ. It often explores multiple possible solⁿ."

Q-2

a Example of a social problem addressed using computational techniques.

Ans Traffic congestion is a major social problem in cities.

A computational solⁿ can use traffic data, GPS information, sensors and ~~algot~~ algorithms to analyse traffic conditions and suggest alternative routes.

Ex - a navigation system can collect information about vehicle speed & road conditions and use algorithms to recommend the shortest or fastest route.

Result: computational techniques can reduce travel time, improve traffic management and help reduce fuel consumption.

b Name & briefly describe two key components of computational thinking.

Ans Decomposition - Breaking a large, complex problem into smaller, more manageable

sub-problems that are easier to understand and solve individually.

- 2 Pattern recognition - Identifying similarities, trends or regularities among and within problems, which helps in predicting outcomes and reusing solutions.

Q-3

a How is computational thinking related to computer science and programming?

ans Computational thinking is a problem solving approach that involves breaking problems with identification and developing algorithm.

C.S. is the broader field that studies computation, algorithms, data, software and information processing.

Programming is the process of implementing an algorithm ~~or~~ using a programming language such as C, C++, Java or python.

Relationship:- Real world problem $\rightarrow$ computational thinking $\rightarrow$ Algorithm $\rightarrow$ problem $\rightarrow$ computer based solⁿ.

Thus, computational thinking provides the problem solving approach, C.S. provides the concepts and principles and

programming implements the solⁿ.

- b Explain computational and informational aspects of a real world problem with an example.

Ans Consider the problem of college attendance management.

The computational aspect deals with the processing and operations performed on the data.

- 1 Calculate attendance percentage
- 2 Determine whether a student has sufficient attendance
- 3 Generate attendance reports.
- 4 Sort students according to attendance
- 5 Identify students below with attendance below a particular limit

The informational aspect deals with what information is collected and stored and communicated.

For eg. student name, roll no., date, etc.

Q-4

- a Compare divide and conquer with mapping to different problems as computational problem using solving approaches. When would one be more suitable than

the other?

Ans Divide and Conquer Mapping to diff. problem

→ Breaks one problem into smaller part. → Relates one problem with another.

→ Suitable for large problems that can be divided independently. → Suitable when a known problem provides an effective solⁿ.

→ eg- Merge sort → eg- Transforming a problem into a known graph problem.

We can use divide & conquer when a problem can naturally be divided into smaller independent or similar problems.

We can also use mapping when the problem resembles another known problem and an existing solⁿ can be adapted.

b Identify a real world problem (eg. traffic congestion) and describe its logical & non logical component.

Ans Let's consider traffic congestion at a busy city intersection.

The logical components that can be represented using rules, data and computational processes:

1. No. of vehicles
2. Traffic signal
3. Vehicles speed
4. Road capacity
5. Traffic flow
6. Peak hours
7. Accident or blockage information

These can be analyzed using algorithms and data.

The non logical part involves human behaviour and factors that may be difficult to represent precisely:

1. Driver impatience
2. Driver behaviour
3. Emergency situation
4. Human mistakes
5. Pedestrian behaviour

A traffic management system can be computationally process logically components, but human behaviour and unpredictable event make some aspects difficult to model perfectly.

Q-5
a

Ans

1

2

3

4

b

Q-5

a Consider a problem like sorting a large dataset. Would a divide & conquer approach be better than an iterative method? Justify your answer.

Ans For sorting a large dataset, a divide and conquer approach can be better, especially when using algorithms such as merge sort.

The reason is:

- 1 The large datasets can be divided into smaller parts.
- 2 Each smaller part can be sorted independently.
- 3 The sorted parts then can be combined.
- 4 The approach is efficient for large dataset.

Eg - Merge sort as follows:

For n elements, merge sort has a time complexity of approximately: $O(n \log n)$

$\therefore$ For large datasets, divide & conquer is often preferable to a simple iterative approach, particularly when efficiency and scalability are important.

b Design a computational approach for a task scheduling system in an organization. Identify the repetitive and non-repetitive aspects and propose a solution.

Ans Consider an ~~organaz~~ organization where different employees have different tasks and deadlines.

Step-1: Input information

The system collects employee ID, Task name, Task priority, dead line, estimated completion time, employ availability, task dependencies.

Step-2: Relative aspects are checking employee availability, assigning tasks, checking deadlines, updating task status, sending reminders, generating daily/weekly reports. These activities can be automated using a computer program.

Step-3: Non repetitive aspects are that some decisions may required a human judgement like handling unexpected emergencies, resolving conflicts between employees, changing priorities because of special circumstances, deciding which task is strategically most important.

Step-4: Proposed computational solⁿ use an algorithm such as input task → check deadlines → check employee availability → assign tasks → monitor progress → send reminders → update schedule.

to:
A computational task scheduling system can be automated repetitive scheduling activity while allowing managers to handle unusual or complex situations. This improves efficiency, time management and productivity.

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